ONLINE SHOPPING APPLICATION

PROJECT ANALYSIS

GROUP 2

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Introduction of the Case

1. Read the case carefully
Understand the project by identifying:

THE PURPOSE OF THE SYSTEM

The Online Shopping Application is a web-based and mobile platform designed to facilitate e-commerce transactions between buyers and sellers. Its core purpose is to provide a seamless digital marketplace where customers can browse products, manage shopping carts, place orders, process payments securely, and track deliveries, all within a single integrated system. The platform also provides sellers with tools to list products, manage inventory, and monitor sales.

THE TARGET USERS

The system is intended for multiple user groups:

- **Customers** – individuals who browse, purchase, and track orders
- **Sellers** – businesses or individuals who list and sell products
- **System Administrator** – personnel who manage accounts, monitor transactions, and ensure platform security.
- **Guest Users** – non-registered visitors who can browse only but not purchase (need to register first before buying)

THE PROJECT SIZE

The Online Shopping Application is considered a large-scale project due to the number of features, modules, and teams required to build and maintain it. The key modules that make up the system are:

- User Registration and Authentication (buyers and sellers)
- Product Catalog and Search Engine
- Shopping Cart and Checkout System
- Payment Gateway Integration / Payment Options (e.g., GCash, PayMaya, credit/debit cards)
- Order Management and Tracking
- Seller Dashboard and Inventory Management
- Reviews and Ratings System

- Admin Control Panel
- Notification System (email/SMS/push)

Because of the many modules involved, the project also requires different teams to work on it such as the frontend, backend, database, security, and quality assurance teams. All of these make the Online Shopping Application a large and complex project to develop.

THE PROJECT COMPLEXITY

The Online Shopping Application is a highly complex project because of the many challenges that need to be addressed during development. The following are the key factors that contribute to its complexity:

- Security the system stores sensitive information like personal details and payment credentials so strong security measures must be applied to protect users.
- Scams and Fraud Prevention since real money is involved the system must be able to detect and prevent fake orders and bogus seller accounts.
- Compatibility the application must work properly across different devices like desktops laptops and mobile phones as well as different browsers.
- Payment Gateway Integration connecting to payment services like GCash and PayMaya is technically difficult and must be implemented carefully.
- Real Time Inventory stock counts must update immediately after every purchase so customers cannot order out of stock items.
- High Traffic during big sale events the system must handle thousands of users at the same time without crashing.
- Multiple User Roles managing different users like customers sellers and admins each with different access levels adds more complexity to the system.

THE EXPECTED DEADLINE

Using the Agile methodology, the Online Shopping Application is expected to be complete within 6 to 7 months. The development is divided into 10 Sprints with each Sprint lasting approximately 2 to 3 weeks.

- Sprint 1 (Week 1 to Week 3) – User registration and login system
- Sprint 2 (Week 4 to Week 6) – Product catalog and search engine
- Sprint 3 (Week 7 to Week 9) – Shopping cart and checkout system
- Sprint 4 (Week 10 to Week 12) – Payment gateway integration
- Sprint 5 (Week 13 to Week 15) – Order management and tracking
- Sprint 6 (Week 16 to Week 18) – Seller dashboard and inventory management
- Sprint 7 (Week 19 to Week 21) – Reviews and ratings system
- Sprint 8 (Week 22 to Week 24) – Admin control panel and notification system

- Sprint 9 (Week 25 to Week 26) – Security testing, bug fixing, and performance optimization
- Sprint 10 (Week 27) – Final testing, quality assurance, and deployment

The total estimated time to complete the Online Shopping Application from planning to deployment is approximately 27 weeks or 6 to 7 months. Maintenance and future updates will continue after the launch based on user feedback.

THE BUDGET LIMITATIONS

The Online Shopping Application has a limited budget that must be carefully managed throughout the development process. To minimize costs the team will prioritize the use of free and open-source tools and frameworks during development.

The budget will be allocated to the following:

- Cloud Hosting the system will be hosted on a cloud platform such as AWS or Google Cloud to ensure that the application is accessible online and can handle multiple users at the same time.
- Payment Gateway Integration connecting to payment services like GCash and PayMaya requires a small transaction fee for every successful payment processed through the system.
- Domain Name a domain name is needed so that users can access the application through a website address without downloading the application and also convenient for desktop users.
- Security budget is needed for security tools and measures to protect the system and its users from data breaches and fraud.
- Third Party Services some features like SMS notifications and email services require third party tools that may have subscription fees.

Due to the limited budget the development team will focus on delivering the most important features first and gradually add more features over time as the budget allows. Unnecessary expenses will be avoided and costs will be monitored regularly throughout the development process to ensure that the project stays within its financial limitations and does not result in financial loss.

THE POSSIBILITY OF CHANGING REQUIREMENTS

The possibility of changing requirements is HIGH. Reasons include:

- User Feedback after each Sprint the users and client may suggest new features or request changes based on what they see and experience.
- Market Trends new features may need to be added to keep up with the competition such as new payment options or delivery tracking features.
- Business Decisions the client may change their mind about certain features or priorities as the project progresses.
- Technology Changes new and better tools or technologies may become available during development that could improve the system and replace the original plan.
- Security Updates new security threats may be discovered during development that require additional security features to be added.

Since the project is using Agile methodology, changes in requirements can easily be accommodated by adding them to the Product Backlog and prioritizing them in the next Sprint.

THE NEED FOR USER/CLIENT FEEDBACK

The need for user and client feedback is HIGH. Regular feedback is essential throughout the development process to ensure the system meets the expectations of both the client and the users. Reasons include:

- User Interface Validation users must regularly review the design and layout to ensure it is easy to navigate and visually appealing.
- Checkout and Payment Flow the buying process must be tested by real users to ensure it is smooth and error free.
- Feature Approval the client must check and approve the features developed after each Sprint to confirm the project is going in the right direction.
- Beta Testing before the full launch a selected group of users will test the system and provide feedback for final adjustments.

Since the project is using Agile methodology, feedback is collected at the end of every Sprint allowing the team to continuously improve the system based on real input.

THE RISKS INVOLVED

The Online Shopping Application involves several risks that must be addressed during and after development. The following are the key risks identified:

- Scams and Fraud since the platform involves real transactions between buyers and sellers there is a risk of fake orders, bogus accounts, and fraudulent activities that could harm users.

- Security Risk the system stores sensitive personal and payment information which makes it a target for hackers and data breaches if proper security measures are not in place.
- Money Handling Risk errors in payment processing or system failures during transactions could result in incorrect charges or lost payments which could damage the trust of users.
- Budget Risk if the development costs exceed the estimated budget the project may be delayed or certain features may need to be removed or simplified.
- Market Competition Risk since there are already many well-known online shopping platforms available today such as Shopee and Lazada it may be difficult to attract users and compete in the market.

Identifying these risks early allows the development team to prepare solutions and preventive measures to reduce their impact on the project.

Project Analysis

2. Identify important project conditions
Look for clues in the case such as:

CLARITY OF REQUIREMENTS

The requirements are partially clear at the beginning of the project. The core features such as user registration, product catalog, shopping cart, and payment system are already identified, which gives a strong foundation for development. However, since the project follows the agile methodology, not all requirements are fully defined in detail at the start. Some features may evolve or be refined as the project progresses. This means the team begins with a general direction but remains flexible to adjust requirements based on feedback and new insights during each Sprint.

LIKELIHOOD OF CLIENT REQUESTING CHANGES

Yes, the client will likely request many changes. This is because agile development encourages continuous improvement and adaptation. After every Sprint, the client reviews the current output and may suggest new features, improvements, or modifications. Additionally, factors such as user feedback, market trends, business decisions, and emerging technologies increase the likelihood of frequent changes throughout the development process.

SYSTEM RISK LEVEL (SAFETY-CRITICAL OR HIGH-RISK)

The system is considered high-risk, but not strictly safety-critical (like medical or aviation systems). The risks are significant because the platform handles financial transactions and sensitive user data.

- The presence of scams and fraud, such as fake sellers or fraudulent orders
- Security threats, including hacking and data breaches involving personal and payment information
- Payment processing errors, which could lead to incorrect charges or financial loss
- Trust issues, where any failure can damage the platform's reputation

Because real money and personal data are involved, strong security, fraud detection, and reliable system performance are critical to minimize these risks.

NEED FOR QUICK DELIVERY

The project requires moderately fast but not rushed delivery. The expected timeline is 6 to 7 months (27 weeks), which is relatively short for a large-scale and complex system. However, the use of Agile with 10 Sprints (each 2–3 weeks) allows the system to be developed and delivered in incremental parts.

- Basic features can be completed and tested early
- Improvements are added gradually
- The system can adapt quickly to feedback

So, while the entire system is not delivered instantly, there is a need for continuous and timely delivery of working features throughout the project.

FREQUENCY OF CUSTOMER FEEDBACK

Yes, customer feedback is frequent and essential. In Agile methodology, feedback is collected at the end of every Sprint.

This frequent feedback is important because:

- Users evaluate the interface and usability

- The checkout and payment process is tested in real scenarios
- The client ensures that features meet expectations
- Beta testing allows real users to identify issues before launch

SYSTEM DEVELOPMENT APPROACH (BUILT IN PARTS)

The system is expected to be built in parts using the agile development approach. This is evident because the project is divided into 10 Sprints, with each Sprint focusing on a specific module such as user registration, product catalog, payment system, and order tracking. In each Sprint, features are developed, tested, and improved step by step before moving on to the next phase. This incremental approach allows the team to manage the project more effectively, identify and fix issues early, and continuously improve the system based on feedback, rather than building the entire system all at once.

This approach allows:

- Easier management of a large and complex system
- Early detection and fixing of issues
- Continuous delivery of usable components
- Flexibility to adjust features based on feedback

Instead of building everything at once, the team gradually constructs the system until all modules are completed and integrated.

Chosen SDLC Model

3. Choose one SDLC model
Select the SDLC model that best fits the case.

RECOMMENDED MODEL: AGILE (Scrum Framework)

After careful analysis of the project characteristics, the Agile methodology specifically implemented through the Scrum framework is selected as the most appropriate SDLC model for the Online Shopping Application. Given the large scale of the project, the high possibility of changing requirements, and the need for regular feedback, Agile is the most suitable approach for this system.

WHAT IS AGILE?

Agile is the mindset and philosophy behind the development of the Online Shopping Application. It values individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a rigid plan. In simple terms Agile is the why and what of the project. It tells the team to build in small parts, welcome changes, and always gather feedback throughout the development process.

WHAT IS SCRUM?

Scrum is the framework or set of techniques used to actually apply the Agile mindset in the project. It organizes the development into time-boxed iterations called Sprints where each Sprint lasting 2 to 3 weeks delivers a working piece of the system. The Online Shopping Application will be developed across 10 Sprints. Scrum gives the team three specific roles and four specific ceremonies to follow throughout the entire development.

SCRUM ROLES

The Scrum framework consists of three roles. The Product Owner represents the client and maintains the Product Backlog. The Scrum Master facilitates ceremonies and ensures the team follows the Agile process. The Development Team is composed of frontend, backend, database, security, and quality assurance members who build the system.

SCRUM CEREMONIES

The Scrum framework follows four ceremonies. Sprint Planning is where the team selects the tasks to complete in the upcoming Sprint. The Daily Standup is a 15 minute daily sync on progress, plans, and blockers. The Sprint Review is where completed features are demonstrated to the client for feedback and approval. Finally the Sprint Retrospective is where the team reflects on improvements before the next Sprint begins.

Justification

4. Explain why your chosen model fits the case

MAIN CHARACTERISTICS OF THE CHOSEN MODEL

Agile, specifically the Scrum framework, isn't just a methodology — it's the most fitting approach for building the Online Shopping Application. In an industry where customer expectations shift overnight and competition is fierce, Scrum's iterative and collaborative nature ensures the application stays relevant, functional, and user-focused from day one. Unlike rigid, linear models that crumble under the weight of change, Scrum thrives in dynamic environments where innovation and responsiveness are non-negotiable — exactly what an e-commerce platform demands.

HOW THOSE CHARACTERISTICS MATCH THE PROJECT NEEDS

- **Speed with Purpose:** Scrum's iterative sprints ensure that working features such as product browsing, checkout, and payment systems are delivered early and often. Stakeholders don't wait months to see progress, they see actual system features in just a few weeks, which builds trust and ensures the system works as expected.
- **Adaptability as a Superpower**: In an online shopping system, requirements can change quickly, such as adding new payment methods like GCash or improving delivery tracking. Scrum embraces these changes, allowing the system to continuously improve and stay competitive.
- **Empowered Teams, Superb Outcomes:** Scrum promotes collaboration among developers, testers, and designers. For this project, different teams work together to ensure that the website or application is functional, secure, and user-friendly, leading to better overall results.
- **Customer-Centric by Design:** The Online Shopping Application heavily depends on user experience. Scrum ensures that customer feedback is collected regularly, especially during sprint reviews, so features like navigation, checkout flow, and product search are easy and effective to use.
- **Risk Reduction Through Transparency:** Scrum practices like daily stand-ups and sprint reviews provide visibility on progress and issues. This helps identify problems early, such as payment errors or system bugs, allowing the team to fix them before they affect users.

ADVANTAGES OF USING THE MODEL

Choosing Scrum for the Online Shopping Application means choosing flexibility, faster delivery, and continuous improvement. The 10-Sprint structure ensures that critical modules like payment gateway integration, order tracking, and user registration are delivered and tested incrementally rather than all at once. The Product Backlog allows the team to easily accommodate changes — such as adding new payment options like GCash or Maya — without disrupting the entire timeline. Regular Sprint Reviews keep the client informed and involved, while built-in testing every Sprint ensures that high-risk components like payment processing and user data security are verified early. Scrum transforms the development process into a continuous cycle of improvement, helping the system remain reliable, user-friendly, and competitive in the e-commerce market.

Comparison with other Models

WHY AGILE IS THE BEST?

Agile is often considered the most effective Software Development Life Cycle (SDLC) model for an online shopping application because the e-commerce industry is highly volatile, competitive, and driven by rapidly changing consumer behavior.

Here is a breakdown of why Agile outperforms traditional models like Waterfall in this specific context:

1. Adaptation to Market Trends

The e-commerce market changes overnight. Whether it is a new payment method (like a new digital wallet), a seasonal sale event (like 11.11 or Black Friday), or a new shipping regulation, Agile allows developers to pivot quickly. Instead of waiting months for a full release, you can integrate new features in **2-4 week sprints**.

2. Continuous User Feedback

Online shopping is all about user experience (UX). Agile relies on constant testing and user input.

- **The Benefit:** You can release a "Minimum Viable Product" (MVP) with basic search and checkout features, then use real customer data to decide if you should build a "Wishlist" feature or a "Recommendation Engine" next.

3. Risk Mitigation and Early Bug Detection

In a Waterfall model, testing happens at the very end. If a critical bug in the payment gateway is found then, it can delay the entire launch.

- **In Agile:** Testing is integrated into every sprint. This ensures that the core functionality—browsing, adding to cart, and paying—is stable and secure from the start.

4. Handling High Complexity

Online shopping apps aren't just storefronts; they are complex ecosystems involving:

- Inventory management systems
- Payment gateway integrations
- Third-party logistics (couriers)
- Customer databases

Agile breaks these complex systems into manageable "User Stories," allowing the team to focus on one functional piece at a time.

COMPARISON TO OTHER MODELS:

1. Agile vs. Waterfall (The Linear Model)

Waterfall is the "old school" approach where you finish one phase completely before moving to the next.

- **Structure:** Waterfall is a straight line; Agile is a continuous circle.
- **Flexibility:** In Waterfall, changing a requirement halfway through is very expensive. In Agile, changes are expected and welcomed.
- **Delivery:** Waterfall delivers the product only at the very end. Agile delivers small, working pieces every few weeks.

2. Agile vs. V-Model (The Validation Model)

The V-Model is a variation of Waterfall that focuses heavily on testing. For every stage of development, there is a corresponding stage of testing.

- **Focus:** The V-Model is about "zero errors" and strict discipline. Agile is about "customer satisfaction" and speed.
- **Suitability:** The V-Model is used for high-stakes software (like medical or flight systems). Agile is used for competitive software (like social media or shopping apps).

3. Agile vs. Spiral Model (The Risk-Focused Model)

The Spiral model is highly sophisticated and focuses on identifying risks early through repeated "loops."

- **Risk:** Spiral is the best for massive, expensive projects where a single mistake could cost millions.
- **Speed:** Agile is much faster. Spiral involves a lot of heavy planning and risk assessment in every loop, which can slow things down.

COMPARISON TABLE:

Feature	Agile	Waterfall	V-Model	Spiral
Flexibility	**Very High**	Low	Low	Medium
Delivery	Frequent small parts	One big final product	One big final product	Progress after every loop
Customer Input	Continuous	Only at start/end	Only at start/end	High (at every loop)
Risk Management	Iterative fixes	Poor	Good	**Excellent**
Complexity	Simple	Simple	Moderate	Very Complex

Conclusion

6. Write your final recommendation

After carefully analyzing the needs, features, and overall requirements of the online shopping application, the chosen SDLC model is Agile Scrum Framework. The main reason for this decision is its strong ability to provide flexibility, continuous improvement, and quick adaptation to user feedback and changing system requirements. Online shopping platforms are dynamic in nature, meaning user preferences, system features, and market trends can change over time. Because of this, using a rigid development model would not be effective. Agile Scrum allows the team to work in smaller iterations or

sprints, making it easier to adjust, improve, and add new features without affecting the entire system.

Another reason for choosing this model is that it promotes better communication and collaboration among team members. Through regular meetings such as daily stand-ups, sprint planning, and reviews, the team can stay aligned, quickly resolve issues, and ensure that everyone is contributing effectively to the project. It also allows stakeholders or users to give feedback regularly, which is very important in developing a system that truly meets user needs.

The expected benefit of using this model is that it enables the development team to deliver functional components of the application step by step, making the system usable at an early stage while still being improved over time. This helps in reducing risks, as problems can be identified and fixed early in the process rather than at the end. Additionally, it improves the overall quality of the system, enhances user satisfaction, and ensures that the final product is reliable, efficient, and user-friendly.